

Computer Architectures

Exam of 25.2.2026 - part I

Question #1B

Let consider the *Hazards* that may arise in a pipelined processor. You are requested to

1. List the different types of Hazards, detailing in particular the different categories of Data Hazards
2. Describe for every type and category an example where the Hazard happens
3. Identify the corresponding data dependency (for Data Hazards, only).

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Question #2B

Let consider the Branch Prediction mechanism based on the Branch Target Buffer (BTB).

You are requested to

1. Describe the architecture of a BTB, detailing the content of the memory it is based on
2. Describe the behavior of a BTB: when it is accessed, which input information items receives, and which output data it produces each time
3. Assuming that the processor uses 32 bit addresses, each instruction is 4 byte wide, and the BTB is composed of 16 entries, provide a table showing the final content of each entry of the BTB and compute the prediction done for each of the branch instructions if
 - The BTB is initially empty (i.e., full of 0s)
 - The following instructions are executed in sequence
 - sub t3, t5, t4 located at the address 0x000A4004
 - beq t2, t5, l1 located at the address 0x000A4008; the branch is taken, and the branch target address is 0x00080024
 - addi t2, t0, #1 located at the address 0x00080024
 - bne t3, t4, l2 located at the address 0x00080028; the branch is taken, and the branch target address is 0x000BB004
 - sll t1, t0, #2 located at the address 0x000BB004
 - sub t1, t0, t3 located at the address 0x000BB008
 - beq t2, t3, l3 located at the address 0x000BB00C; the branch is not taken
 - beq t2, t5, l4 located at the address 0x000BB010; the branch is taken, and the branch target address is 0x000A4004
 - add t1, t2, t3 located at the address 0x000A4004
 - beq t2, t5, l1 located at the address 0x000A4008; the branch is taken, and the branch target address is 0x00080024.

Index	Final BTB Content
0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	